React Overview

In this lesson, we will delve into an overview of React, one of the prominent libraries for building modern Single Page Applications (SPAs).

What is React?

React is specifically tailored for creating SPAs, but you might be wondering: what distinguishes a single page application from a traditional application?

Traditional Applications vs. Single Page Applications

In a **traditional application**, each user action triggers a full HTML page load. For instance, when you engage with any element on the page, it prompts a new request to the server, resulting in a full page reload each time.



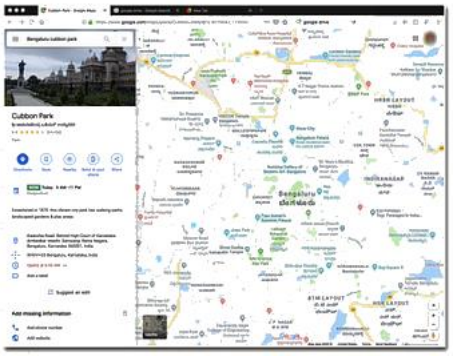
Single Page Applications, on the other hand, are made up of just one page. Based on user interactions, the application page receives partial updates instead of full page reloads. The SPA can also communicate with the backend server, sending and retrieving data using REST APIs, which can then be used to populate the application's interface.



Examples of SPAs

You may wish to see concrete examples of SPAs to understand them better. Popular services like Google Maps, Yahoo Maps, Gmail, or Yahoo Mail are all SPAs. These applications make partial updates depending on the user's actions, like searching for a location or clicking an email, without needing a full-page refresh.

Maps



Email



React's Role in SPAs

React plays an essential role in building modern SPAs. It allows for partial page updates and communicates with backend services using REST APIs. In our upcoming lessons, we will actually build a full-stack application, using React for the frontend and Spring Boot for the backend, all working seamlessly together.



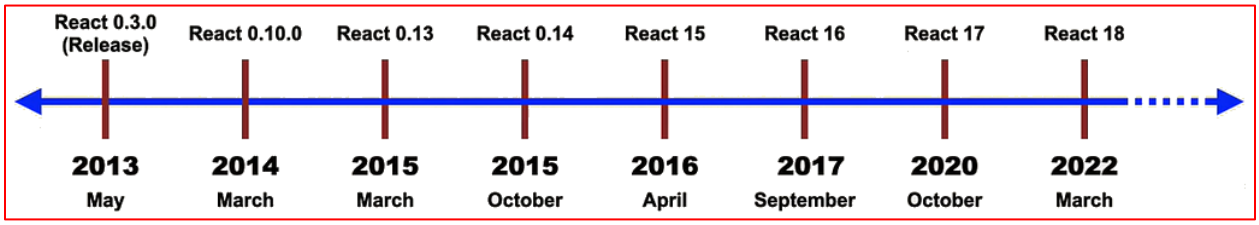
Who Uses React?

You may wonder about the real-world adoption of React. Companies globally utilize React in their projects. Some well-known users include Dropbox, WhatsApp, and Facebook (React's original developer). You can explore more about projects using React in the real world at madewithreact.com.



React's History

React has come a long way since its initial release as version 0.3.0 in May 2013. New versions continue to be launched, and we're currently at the version shown on your screen, but expect more updates in the future.



Why Not DIY with Vanilla JavaScript?

Could you build an SPA using vanilla JavaScript, jQuery, and Ajax? While it might be suitable for small hobby apps, you'd find challenges in implementing common features like data binding, especially in complex, real-time enterprise applications involving multiple developers.

That's where frameworks and libraries like React, Angular, Vue, and others come into play. These tools handle the heavy lifting and low-level intricacies, enabling developers to concentrate on building the app itself.